FSDS Research Progress Report

Cross-Lingual Multi-Hop Machine Unlearning

Researcher: Mai Anh Pham | **Report:** 01 | **Period:** 22 October–3 November 2026

Stage: v4 feasibility pilot complete; competence gate failed; corrected sampler rerun pending

github.com/aCoderChild/fsds

1 Research Problem and Motivation

Subfield: Cross-lingual Transfer of Machine Unlearning in Multi-hop Question Answering.

Main idea: I investigate whether unlearning an intermediate factual relationship using English-only data affects the model's ability to answer related multi-hop questions in other languages: French, German, Hindi, and Thai. I distinguish direct recall of the targeted relationship from the ability to reach its downstream final answer. This distinction is motivated by prior work on knowledge-edit propagation and multilingual forgetting [5, 9].

Research goal: An empirical evaluation of cross-lingual downstream forgetting, subject to verifying that the model learned the relevant facts and that the final answers depend on the targeted relationships [10, 7].

2 Research Question and Hypothesis

Research question. Does English-only unlearning of an intermediate factual relationship change the accuracy of downstream multi-hop answers in other languages?

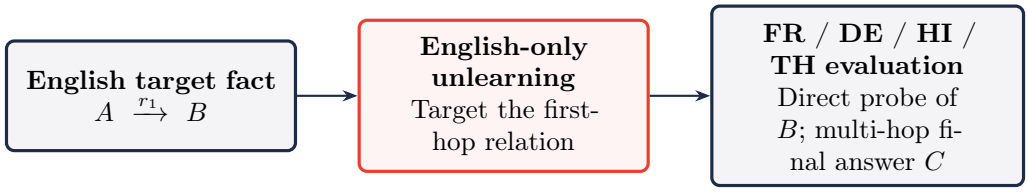


Figure 1: Research intuition: target the relationship $A \to B$ using English-only unlearning, then test direct recall of B and the downstream answer C in other languages. A correct final answer alone does not prove recovery of B [7].

Hypothesis. English-only unlearning can reduce direct factual recall in other languages, but its effect on multi-hop final-answer accuracy may vary with language and reasoning depth. This is a testable prediction, not an assumed outcome [9, 13].

3 Literature Review

3.1 Cross-lingual Machine Unlearning

Xiang et al. [9] investigate whether knowledge unlearned in one language remains accessible in other languages. Their findings show that cross-lingual unlearning transfer varies considerably across languages and that some forgotten knowledge can be recovered.

Similarly, Waseem and Fawaz [10] demonstrate that cross-lingual forgetting can be misinterpreted when the model never learned the target knowledge in the evaluation language. They emphasize the importance of comparing the base model, the fine-tuned model, and the unlearned model, while also controlling for general performance degradation.

These studies establish that **cross-lingual forgetting is variable and needs pre-learning controls** [9, 10]. However, they do not directly establish how forgetting an intermediate fact affects the final answer of a multi-hop reasoning question in another language.

3.2 Multi-hop Reasoning and Knowledge Unlearning

MQuAKE [5] evaluates whether editing a factual relationship leads to the expected changes in downstream multi-hop answers. It shows that successfully updating an individual fact does not necessarily produce

consistent changes in questions that depend on that fact.

R-TOFU [12] extends unlearning evaluation to reasoning traces and demonstrates that information may remain in intermediate reasoning steps even when final answers appear to be forgotten.

Qian et al. [13] further study how unlearned knowledge can remain accessible through different multi-hop reasoning paths and recovery attacks.

These works highlight the importance of evaluating knowledge beyond direct factual questions [12, 13]. However, they do not jointly isolate the cross-lingual effects of unlearning a specific intermediate relationship on downstream multi-hop answers.

3.3 Controlled Knowledge and Reasoning Shortcuts

TOFU [6] introduces fictional author profiles to study machine unlearning under more controlled conditions. Using fictional knowledge reduces interference from information that a model may already have learned during pretraining.

However, fictional data alone does not guarantee that a model actually performs multi-hop reasoning. The model may memorize an association between the starting entity and the final answer.

SOCRATES [7] addresses this problem by constructing multi-hop questions that reduce opportunities for shortcut-based answering. CLUTRR [8] also provides controlled relational reasoning tasks that test generalization to unseen combinations of relations.

These studies motivate using controlled facts and shortcut-aware questions to test whether final answers actually depend on the intended intermediate relationships [7, 8].

3.4 Research Gap

Existing research has investigated cross-lingual unlearning, multi-hop knowledge leakage, and shortcut-aware reasoning evaluation.

The intended work focuses on their *intersection*:

Does unlearning an intermediate factual relationship using English-only data affect the model's ability to produce correct final answers to multi-hop questions in other languages?

I plan to evaluate downstream multi-hop accuracy after English-only unlearning, while controlling for prior knowledge, insufficient learning, reasoning shortcuts, and collateral damage on retained facts [10, 14].

4 Experiment Design

4.1 Datasets

I plan to use SynthWorlds-SM, the synthetic-mapped version of SynthWorlds, which provides a corpus of synthetic entities and relationships alongside existing multi-hop question-answering tasks [1]. The dataset is designed to distinguish reasoning performance from advantages due to pretrained world knowledge.

Dataset: SynthWorlds on Hugging Face.

1. **Dataset preparation:** Reuse SynthWorlds-SM factual triples and multi-hop questions without inventing new facts [1]. Audit whether every selected target relationship lies on a valid dependency path to its final answer.
2. **Translation:** Translate the selected questions and factual statements into English (EN), French (FR), German (DE), Hindi (HI), and Thai (TH); manually audit a sample for entity and answer consistency.
3. **Dataset splitting:** Split by factual chain or connected entity group into forget, retain, and holdout sets to prevent leakage.
4. **Knowledge acquisition:** Fine-tune the model on factual relationships in all five languages using SFT.
5. **Pre-unlearning verification:** Verify direct factual recall and multi-hop final-answer accuracy in each language.
6. **English-only unlearning:** Unlearn selected intermediate factual relationships using English-only data.
7. **Cross-lingual evaluation:** Compare multi-hop final-answer accuracy before and after unlearning across

all five languages.

4.2 Models and Knowledge Acquisition

1. **Model selection:** Use Llama 3.2 1B-Instruct for the feasibility pilot and Llama 3.1 8B-Instruct for the main study [3, 2].
2. **Motivation:** Test whether language-dependent model competence is associated with English-only forgetting transfer. Meta lists all five study languages among Llama 3.1's supported languages [2].
3. **Knowledge acquisition:** Use supervised fine-tuning (SFT) to teach factual relationships from SynthWorlds-SM in five languages: English (EN), French (FR), German (DE), Hindi (HI), and Thai (TH).
4. **Reasoning preparation:** Exclude evaluation multi-hop QA pairs and direct start-to-final-answer shortcuts from factual SFT. If necessary, train reasoning composition on disjoint chains [7, 8].

4.3 Pre-Unlearning Verification

1. **Baseline evaluation:** Evaluate the pretrained model to identify any existing factual knowledge.
2. **Knowledge verification:** After SFT, evaluate the model in all five languages using:
   - Direct factual recall.
   - Multi-hop final-answer accuracy.
   - Alternative question phrasings.
3. **Case selection:** Identify factual chains answered correctly before unlearning and record the eligible cases for each language and reasoning depth.
4. **Evaluation control:** Compare pre- and post-unlearning performance on the same cases to avoid treating unlearned knowledge as forgotten knowledge [10].

4.4 English-Only Unlearning

1. **Unlearning target:** Target the first factual relationship in selected multi-hop chains using English-only QA pairs.
2. **Unlearning methods:** Compare three methods:
   - Gradient Ascent (GA).
   - Gradient Difference (GD).
   - Negative Preference Optimization (NPO) [11].
3. **Reference models:** Compare with pretrained, full-SFT, and retain-only checkpoints to distinguish forgetting from weak acquisition and general degradation [10, 6].
4. **Utility preservation:** Monitor retain-set accuracy and generation quality to distinguish targeted forgetting from general model degradation [11, 14].

4.5 Cross-Lingual Multi-hop Evaluation

1. **Evaluation languages:** English (EN), French (FR), German (DE), Hindi (HI), and Thai (TH).
2. **Evaluation metrics:**
   - **Direct recall:** Accuracy on targeted intermediate facts.
   - **Multi-hop accuracy:** Accuracy of final answers to multi-hop questions involving the targeted facts (primary metric).
   - **Knowledge retention:** Accuracy on unrelated factual chains.
3. **Cross-lingual transfer:** Compare pre- and post-unlearning accuracy on matched factual chains in each language.
4. **Selectivity:** Compare changes in forget-set and retain-set performance to identify unintended degradation [6, 14].

4.6 Controlling for Reasoning Shortcuts

1. **Prevent answer memorization:** Exclude exact evaluation multi-hop QA pairs from SFT.
2. **Shortcut detection:** Check for direct associations between starting entities and final answers.
3. **Dependency verification:** Where supported by the dataset, test alternative questions or factual paths using existing verified relationships [5, 7].
4. **Interpretation:** Distinguish correct downstream answers from verified recovery of intermediate knowledge.

4.7 Analysis Plan

I will evaluate whether English-only unlearning of intermediate factual relationships affects multi-hop question answering across five languages: English (EN), French (FR), German (DE), Hindi (HI), and Thai (TH).

Evaluation Metrics

1. **Pre-unlearning competence:** Measure direct factual recall and multi-hop final-answer accuracy after SFT. Record cases answered correctly before unlearning [10].
2. **Direct forgetting:** Measure the reduction in recall of targeted intermediate facts across languages [9].
3. **Multi-hop forgetting (Primary):** Measure the change in final-answer accuracy on multi-hop questions involving targeted intermediate facts [5].
4. **Downstream answer persistence:** Measure how often the final answer remains correct when direct recall of the targeted intermediate fact fails. This is a proposed case-level metric; it is not evidence of recovered reasoning traces [7].
5. **Knowledge retention:** Measure accuracy on unrelated factual chains before and after unlearning [6].
6. **Cross-lingual persistence:** Consider the Knowledge Persistence Score (KPS) for consistency of forgetting across language pairs, using the original definition of Hwang et al. [4].
7. **Reliability and validity:** Compare methods across languages and hop depths. Report generation failures, repeated seeds, and paired confidence intervals. Examine possible reasoning shortcuts [7].

Metric Definitions

Let $\mathrm{Acc}_{\mathrm{pre}}$ and $\mathrm{Acc}_{\mathrm{post}}$ denote accuracy before and after unlearning.

The change in multi-hop accuracy is:

$$\Delta_{\mathrm{MH}} = \mathrm{Acc}_{\mathrm{post,MH}} - \mathrm{Acc}_{\mathrm{pre,MH}}.$$

A negative value indicates reduced downstream multi-hop performance.

Direct forgetting is measured as:

$$F_{\mathrm{direct}} = \mathrm{Acc}_{\mathrm{pre,direct}} - \mathrm{Acc}_{\mathrm{post,direct}}.$$

A positive value indicates reduced direct recall.

For downstream answer persistence, let B denote the targeted intermediate fact and C the multi-hop final answer.

For each language and split, first restrict to cases where full SFT answered both the direct and multi-hop probes correctly. Within that pre-unlearning competent set, I will calculate:

$$\mathrm{DAP} = \frac{N(B_{\mathrm{pre}} = \mathrm{correct}, C_{\mathrm{pre}} = \mathrm{correct}, B_{\mathrm{post}} = \mathrm{wrong}, C_{\mathrm{post}} = \mathrm{correct})}{N(B_{\mathrm{pre}} = \mathrm{correct}, C_{\mathrm{pre}} = \mathrm{correct}, B_{\mathrm{post}} = \mathrm{wrong})}.$$

DAP measures the proportion of direct-recall failures for which the downstream answer remains correct. It does not establish recovery of the intermediate fact. If no direct-recall failures occur, DAP is undefined and will be reported as N/A.

KPS, if reported, will follow its published definition [4] and will not be conflated with the proposed downstream answer persistence metric.

Proposed Results Tables

Table 1. Pre-unlearning Knowledge Verification

Language	Total	Direct (%)	MH (%)	Eligible
English (EN)	–	–	–	–
French (FR)	–	–	–	–
German (DE)	–	–	–	–
Hindi (HI)	–	–	–	–
Thai (TH)	–	–	–	–

Table 1: Pre-unlearning factual recall and multi-hop (MH) accuracy after SFT. Eligible cases are those answered correctly on both tasks before unlearning.

The number of eligible cases will be reported for each language and reasoning depth.

I will report results on the full evaluation set and separately on eligible cases to distinguish forgetting from incomplete knowledge acquisition.

Table 2. Main Cross-Lingual Unlearning Results

Method	Lang.	N	Direct (%)	MH (%)	Retain Δ pp	Invalid (%)
GA	EN	–	–	–	–	–
	FR	–	–	–	–	–
	DE	–	–	–	–	–
	HI	–	–	–	–	–
	TH	–	–	–	–	–
GD	EN	–	–	–	–	–
	FR	–	–	–	–	–
	DE	–	–	–	–	–
	HI	–	–	–	–	–
	TH	–	–	–	–	–
NPO	EN	–	–	–	–	–
	FR	–	–	–	–	–
	DE	–	–	–	–	–
	HI	–	–	–	–	–
	TH	–	–	–	–	–

Table 2: Post-unlearning accuracy on previously correct forget-set cases. N is the number of eligible cases. Direct and MH are post-unlearning accuracies. Retain Δ is the percentage-point change (post minus pre) on the unrelated retain set. Invalid denotes malformed or repetitive outputs.

Table 3. Intermediate-Fact and Final-Answer Outcomes

The main outcome of interest is **B fail, C pass**, where direct factual recall fails but the downstream answer remains correct.

I will distinguish this outcome from verified recovery of intermediate knowledge.

Table 4. Effect of Reasoning Depth

Statistical Analysis

1. Repeat unlearning experiments using at least three random seeds.
2. Report mean accuracy and standard deviation across seeds.
3. Calculate paired 95% bootstrap confidence intervals over factual chains, preserving matched case identities before and after unlearning.

Method	Lang.	B fail C pass	B fail C fail	B pass C pass	B pass C fail
GA	EN	–	–	–	–
	FR	–	–	–	–
	DE	–	–	–	–
	HI	–	–	–	–
	TH	–	–	–	–
GD	EN	–	–	–	–
	FR	–	–	–	–
	DE	–	–	–	–
	HI	–	–	–	–
	TH	–	–	–	–
NPO	EN	–	–	–	–
	FR	–	–	–	–
	DE	–	–	–	–
	HI	–	–	–	–
	TH	–	–	–	–

Table 3: Paired direct and multi-hop outcomes after unlearning. B denotes the targeted intermediate fact and C the final answer. Each cell reports the percentage of eligible cases. The four outcomes sum to 100% within each method-language group.

Method	Language	2-hop	3-hop	4-hop
GA	EN	–	–	–
GD	EN	–	–	–
NPO	EN	–	–	–
GA	FR	–	–	–
GD	FR	–	–	–
NPO	FR	–	–	–
GA	DE	–	–	–
GD	DE	–	–	–
NPO	DE	–	–	–
GA	HI	–	–	–
GD	HI	–	–	–
NPO	HI	–	–	–
GA	TH	–	–	–
GD	TH	–	–	–
NPO	TH	–	–	–

Table 4: Change in multi-hop final-answer accuracy (Δ_{MH}, percentage points) after English-only unlearning, grouped by language and reasoning depth. Results are reported only for hop depths supported by sufficient eligible cases.

4. Compare changes between forget and retain sets to assess selective forgetting.
5. Compare outcomes across languages, reasoning depths, and unlearning methods.
6. Inspect incorrect, malformed, and repetitive responses to identify generation collapse and possible reasoning shortcuts.

The main analysis will ask whether changes in direct recall of the targeted intermediate fact accompany downstream multi-hop accuracy changes in other languages, without comparable degradation on unrelated factual chains [14].

Feasibility-pilot result (2026-11-02)

The evaluation-disjoint SynthWorlds-SM pilot completed five-language LoRA SFT and a baseline-only evaluation in the Kaggle Notebook Editor using Llama 3.2 1B-Instruct. The SFT used 1,655 rows (331 per language): 765 forget-split atomic-fact rows, 390 retain-split rows, and 500 rows from 50 additional training-only chains. The retain-only reference used 890 retain and training-only rows. Both runs used three epochs and seed 42. The 129 evaluation chains were grouped into 77 forget, 26 retain, and 26 holdout cases.

On the forget split, SFT direct-recall and multi-hop exact-match accuracy were:

	EN	FR	DE	HI	TH
Direct recall (%)	88.3	84.4	77.9	83.1	83.1
Multi-hop answer (%)	10.4	9.1	18.2	14.3	15.6

The evaluator produced 3,870 predictions across the pretrained, full-SFT, and retain-only checkpoints; it recorded zero generation failures and zero output-invalid flags. The holdout accuracy was zero because its facts were excluded from training; this is a split-control result, not evidence of forgetting. Retain results overlap retain-composition training and are training-overlap controls.

Only eight of the 77 English forget cases were SFT-precompetent on both direct and multi-hop probes. The paired English-plus-language counts were FR 2 cases/2 entity groups, DE 4/3, HI 4/3, and TH 3/3. Each language falls below the predeclared pilot feasibility floor of 15 cases and 8 groups. At that point, the baseline gate blocked the standard unlearning path and post-unlearning evaluation. The exploratory override run on 2026-11-03 is reported separately below. The baseline alone does not answer the cross-language unlearning hypothesis. Translations were reviewed AI-only; fluent human review was not performed. The exact forget multi-hop prompts were excluded from SFT, but their atomic supporting facts were included, so this tests held-out composition prompts over trained facts rather than unseen-fact generalization.

Follow-up diagnostic update (2026-11-03)

The v4 composition corpus increased five-language SFT to 3,355 rows (671 per language), adding 85 source-verified Type-B training-only chains. Kaggle run manifests record completion of full SFT, a retain-only reference, and baseline evaluation on the same frozen 129-case evaluation set. The baseline evaluation covered 3,870 predictions; the exploratory post-unlearning evaluation covered 7,740. Both archives report zero generation failures and zero deterministic output-invalid flags. Archive hashes and row counts were verified, and rescoring the saved predictions with the current evaluator reproduced the scored predictions, summaries, and contrast files byte-for-byte.

The predeclared bilingual-competence gate still failed. Among 77 forget cases, 15 were English SFT-precompetent under the direct-plus-full-multi-hop selection rule. Matched English/non-English precompetent cases and leakage-isolated groups were FR 10/6, DE 9/5, HI 9/6, and TH 11/7; the floor was 15 cases and 8 groups for each language. An explicitly labeled exploratory override then ran GA, GD, and NPO on the 15 English first-hop forget rows, with no holdout rows. The primary contrast is the target-language multi-hop change minus the English multi-hop change (percentage points), computed over paired precompetent cases and bootstrapped by entity group:

Language	Cases/groups	GA	GD	NPO
FR	10/6	+10.0 [0.0, 50.0]	+10.0 [0.0, 50.0]	+10.0 [0.0, 50.0]
DE	9/5	+22.2 [0.0, 80.0]	+11.1 [-33.3, 80.0]	+11.1 [-33.3, 80.0]
HI	9/6	0.0 [-33.3, 33.3]	+11.1 [0.0, 50.0]	+11.1 [0.0, 50.0]
TH	11/7	+9.1 [-28.6, 57.1]	+9.1 [-42.9, 71.4]	+9.1 [-27.3, 57.1]

All intervals include zero and are wide. These values do not provide reliable evidence for or against a cross-language effect. A code audit found that the archived GD/NPO runs reused only 15 of 26 available English retain rows because their sampler restarted at the beginning each epoch. GA does not use a retain regularizer. The archived GD/NPO estimates apply only to the earlier sampler; a corrected Kaggle rerun is still required. The corrected sampler cycles through every retain row before reuse; fixture tests pass. This run is one seed on a 1B model, and all translations are AI-only reviewed without fluent-human review. It remains exploratory and does not answer the hypothesis. Reproducibility artifacts are in `reports/hypothesis_diagnostics/r1_synthworlds_v4_exploratory_2026-11-03/`.

5 Ongoing Work

Work item	Current state	Status
Dataset feasibility	Built a leakage-isolated 129-case pilot and a 3,355-row five-language SFT corpus. The 1B baseline leaves only 9–11 paired precompetent cases per target language, below the predeclared floor.	Expand eligibility
Experimental design	Ran five-language SFT, baseline evaluation, and labeled exploratory GA/GD/NPO. The competence gate failed; the corrected GD/NPO sampler still needs its Kaggle rerun.	Rerun pending
Model selection	Llama 3.2 1B ran on a Kaggle T4. Llama 3.1 8B remains the proposed main-study model; its runtime and multilingual competence have not been tested.	Pilot verified
Evaluation controls	The competence gate failed. Exploratory post-evaluation exists; retain overlap and unseen-fact limitations remain as described above.	Exploratory

6 Issues / Blockers and Support Needed

- **Dataset suitability and competence:** The pilot's bilingual precompetent subset is too small for interpretable unlearning. The evaluation translations are AI-only reviewed, not fluent-human validated. I need feedback on expanding the eligible set and validating the target relations and translations [1].
- **Experimental validity:** Review of the evaluation protocol, baseline models, and metrics.
- **Computational resources:** The 1B feasibility run completed on a Kaggle T4. The planned Llama 3.1 8B main experiment, repeated seeds, and multiple unlearning methods still require a compute-feasibility assessment.

  Support needed: Advice on computational feasibility and GPU resources for the main experiments.
- **Research contribution:** Feedback on the novelty of my research question and the results required for a publishable contribution.

7 Plan Until Next Report

☐ **Pilot follow-up:** Expand the leakage-isolated set and improve SFT-precompetent bilingual coverage without changing the eligibility rule after seeing results.

☐ **Translation validation:** Obtain fluent-speaker review of the AI-only French, German, Hindi, and Thai translations, then recheck entity and target-fact consistency.

☐ **Experimental protocol:** Obtain supervisor feedback on the current leakage-isolated split, competence threshold, and paired evaluation plan before expanding the pilot into a main study.

☐ **Next experiment stage:** Upload the corrected versioned package and rerun the sampler-fixed GD/NPO exploratory stage in the Kaggle Notebook Editor. Keep the gate failure and small cohort explicit. Expand the dataset and obtain fluent-speaker translation review before a powered study; run confirmatory unlearning only after its preregistered competence gate passes.

8 Targeted Journal/Conference Submission Information

Venue	Deadline	Current Stage	Notes
ACL 2027	04/01/2027	Experiment Design	Potential target if experiments and analysis are completed in time.
EMNLP 2027	To be announced	Experiment Design	Primary target for a controlled empirical study of cross-lingual multi-hop unlearning.

9 References

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Supervisor's Comment and Check

Supervisor	*To be confirmed*
Date Reviewed	
Comments	

Co-supervisor 1	*To be confirmed*
Date Reviewed	
Comments	

Co-supervisor 2	*To be confirmed*
Date Reviewed	
Comments	